

NORYL™ RESIN 731S

REGION ASIA

DESCRIPTION

NORYL™ 731S resin is a non-reinforced blend of Polyphenylene ether (PPE) + high impact polystyrene (HIPS). This injection moldable resin is FC EU, FDA food contact compliant and has NSF/ANSI 61, ACS, WRAS, and KTW/W270 listings for selected colors and manufacturing locations. NORYL 731S resin exhibits good surface appearance, high ductility, low moisture absorption, dimensional and hydrolytic stability. NORYL 731S resin is an excellent candidate for a variety of water management applications such as valves, filtration components, and water meter internals.

TYPICAL PROPERTY VALUES

Revision 20210812

PROPERTIES	TYPICAL VALUES	UNITS	TEST METHODS
MECHANICAL			
Tensile Stress, yld, Type I, 50 mm/min	58	MPa	ASTM D638
Tensile Strain, brk, Type I, 50 mm/min	30	%	ASTM D638
Flexural Stress, yld, 1.3 mm/min, 50 mm span	93	MPa	ASTM D790
Flexural Modulus, 1.3 mm/min, 50 mm span	2420	MPa	ASTM D790
IMPACT			
Izod Impact, notched, 23°C	208	J/m	ASTM D256
Izod Impact, notched, -40°C	133	J/m	ASTM D256
THERMAL			
HDT, 1.82 MPa, 3.2mm, unannealed	117	°C	ASTM D648
HDT, 1.82 MPa, 6.4 mm, unannealed	126	°C	ASTM D648
CTE, -40°C to 60°C, flow	1.3E-04	1 /°C	ASTM E831
Relative Temp Index, Elec ⁽¹⁾	105	°C	UL 746B
Relative Temp Index, Mech w/impact ⁽¹⁾	90	°C	UL 746B
Relative Temp Index, Mech w/o impact ⁽¹⁾	105	°C	UL 746B
PHYSICAL			
Specific Gravity	1.08	-	ASTM D792
Water Absorption, (23°C/24hrs)	0.06	%	ASTM D570
Mold Shrinkage, flow, 3.2 mm	0.5 – 0.7	%	SABIC method
ELECTRICAL			
Dissipation Factor, 50/60 Hz	2.65	-	ASTM D150
Comparative Tracking Index (UL) {PLC}	2	PLC Code	UL 746A
High Amp Arc Ignition (HAI), PLC 2	≥1.5	mm	UL 746A
High Amp Arc Ignition (HAI), PLC 3	≥3	mm	UL 746A
Hot-Wire Ignition (HWI), PLC 1	≥1.5	mm	UL 746A
High Voltage Arc Track Rate {PLC}	4	PLC Code	UL 746A
Arc Resistance, Tungsten {PLC}	6	PLC Code	ASTM D495
FLAME CHARACTERISTICS ⁽¹⁾			
UL Yellow Card Link	E207780-228540	-	-
UL Recognized, 94HB Flame Class Rating	≥1.5	mm	UL 94
INJECTION MOLDING			
Drying Temperature	105 – 110	°C	
Drying Time	2 – 4	Hrs	

PROPERTIES	TYPICAL VALUES	UNITS	TEST METHODS
Drying Time (Cumulative)	8	Hrs	
Melt Temperature	280 – 310	°C	
Mold Temperature	70 – 95	°C	

(1) UL Ratings shown on the technical datasheet might not cover the full range of thicknesses and colors. For details, please see the UL Yellow Card.

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